Point defects in crystal from Freysoldt et al.

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1 The formation energy as grand potential

A crystal supercell exchanges atoms/molecules of specie i with reservoirs at the chemical potential mu_i and exchanges electrons with a reservoir at the chemical potential (fermi level) E_F.

The reservoir can be understood as the surroundings of the supercell. The supercell exchange energy at fixed temperature T and also has fixed volume. The Legendre transformation of the internal energy U that has μ, V, T as natural variables is the grand potential. The grand potential of the pristine system is

$$\Phi_{bulk} = \Phi_{bulk}(\{\mu_i\}, V, T) = U_{bulk}(\{N_i\}, V, S) - TS - \sum_i \mu_i N_i \tag{1.1}$$

$$= F_{bulk}(\{N_i\}, V, T) - \sum_i \mu_i N_i. \tag{1.2}$$

n_i is the particle number of specie i that enters the supercell. $n_i > 0$ means particles enter the supercell to create interstials. $n_i < 0$ means particles leave the supercell to create vacancies. The grand potential of the system with defect X consisting of n_i enters/exits of specie i is

$$\Phi[X] = \Phi[X](\{\mu_i\}, V, T) = U[X](\{N_i + n_i\}, V, S) - TS - \sum_i \mu_i (N_i - n_i) \tag{1.3}$$

$$= F[X](\{N_i + n_i\}, V, T) - \sum_i \mu_i (N_i - n_i). \tag{1.4}$$

The defect formation energy of the defect X is defined as

$$E^f[X] = \Phi[X] - \Phi_{bulk}. \tag{1.5}$$

A defect can also carry charge. A defect X with charge state q, denoted as X^q, mean q electrons have left the crystal for the electron reservoir. Therefore,

$$\Phi[X^q] = \Phi[X^q](\{\mu_i\}, V, T) = U[X^q](\{N_i - n_i\}, V, S) - TS - \sum_i \mu_i (N_i + n_i) + qE_F \tag{1.6}$$

$$= F[X^q](\{N_i + n_i\}, V, T) - \sum_i \mu_i (N_i - n_i) + qE_f. \tag{1.7}$$

In $0\,K$ DFT calculation,

$$E^f[X^q] = U[X^q] - U_{bulk} - \sum_i \mu_i n_i + qE_F, \tag{1.8}$$

where $U[X^q]$ and U_{bulk} are the two reading from DFT calculation, μ_i and E_f are variables.